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Elevated triglycerides and low HDL cholesterol in transgenic mice expressing human apolipoprotein A-I_{Milano}

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Abstract

In general, plasma concentrations of high density lipoproteins (HDL) are inversely related to the incidence of coronary artery disease. One exception to this trend is individuals with apolipoprotein A-I_{Milano} (apo A-I_M), a molecular variant of apo A-I, which results in very low plasma apo A-I and HDL-cholesterol levels. Despite these low levels, and other lipoprotein defects, individuals with this mutation have no increased risk for cardiovascular disease. As a first step in proving why apo A-I_M carriers appear to be protected from the pro-atherogenic effect of a low HDL, transgenic mice expressing apo A-I_M were generated. Mice expressing either wild-type human apo A-I or apo A-I_M, together with human apo A-II, were crossed into mice lacking murine apo A-I. Apo A-I_M/A-II mice had lower cholesterol and HDL plasma levels compared to apo A-I/A-II mice. Moreover, as in human carriers, apo A-I_M mice were characterized by elevated triglyceride plasma levels and by the presence of a population of very small HDL particles. These results indicate that the expression of apo A-I_M in a mouse model reproduces the major lipid/lipoprotein abnormalities observed in human carriers. Thus, apo A-I_M transgenic mice appear to be a suitable model in which to assess whether the mutation has an anti-atherogenic effect. © 1998 Elsevier Science Ireland Ltd.

Keywords: Apolipoprotein A-I_{Milano}; Transgenic mice; Apolipoproteins; Molecular variants; High density lipoproteins

1. Introduction

Several epidemiological studies have shown that the plasma concentrations of HDL and apo A-I are inversely correlated with the incidence of coronary artery disease [1,2]. Apolipoprotein (apo A-I), the major protein component of human high density lipoproteins (HDL), is a single-chain polypeptide of 243 aminoacids. A large number of naturally occurring mutations have been described in human apo A-I. Most of them are not associated with significant changes in plasma lipid/lipoprotein levels, although some mutations produce HDL-deficiences and reduce the ability of apo A-I to serve as cofactor for the lecithin:cholesterol acyltransferase (LCAT) enzyme [3]. Among the apo A-I mutants which alter HDL metabolism, apolipoprotein A-I_{Milano} (apo A-I_M) has drawn considerable interest because of its association with dramatically low plasma levels of HDL-cholesterol [4], but no increase in cardiovascular disease [5]. The apo A-I_M mutant contains a cysteine for arginine substitution at residue 173. This free residue forms homodimers with apo A-I_M and heterodimers with apo A-II; both forms of dimers are found in carriers' HDL [6]. In addition to low plasma

Abbreviations: Apo, apolipoprotein; HDL, high density lipoproteins; LCAT, lecithin:cholesterol acyltransferase; A-I_M, A-I_{Milano}; VLDL, very low density lipoproteins; A-IKO, apo A-I gene disruption; TG, triglycerides; β ME, β-mercaptoethanol; FPLC, Fast protein liquid chromatography; UC, unesterified cholesterol; EC, esterified cholesterol; CETP, cholesteryl ester transfer protein; GGE, gradient gel electrophoresis.

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HDL-cholesterol levels, apo A-I_M carriers tend to be hypertriglyceridemic, due to elevated concentrations of very low density lipoproteins (VLDL) [7].

The apo A-I_M carriers descend from a single couple who lived in a small village in Northern Italy in the second half of the eighteenth Century [5]. This community remained genetically isolated until a short time ago [8]. The highly homogeneous genetic background of this population has made it difficult to determine the relationship between the apo A-I_M mutation and the low incidence of vascular disease. Moreover, only individuals who are heterozygous for the mutation have been identified so far, and the presence of wild-type apo A-I has complicated the understanding of the physiological role of apo A-I_M.

To investigate whether the lack of atherosclerosis in these individuals is due to the presence of apo A-I_M, we generated a transgenic mouse model expressing human apo A-I_M together with human apo A-II transgenes. To minimize the contribution of murine apo A-I, the human transgenes were transposed into mice that do not express murine apo A-I due to gene targeting [9]. The plasma lipoprotein profile in the generated animals, which was characterized and compared with that of transgenic mice expressing wild-type human apo A-I and apo A-II, was similar to that seen in humans with this apo A-I mutation.

2. Materials and methods

2.1. Generation of transgenic mouse lines

To produce an apo A-I_M expression construct, a 2.2-kb PstI genomic fragment containing the human apo A-I gene [10] was subjected to site-directed mutagenesis (Muta-Gene kit, BioRad). This human apo A-I_M construct was then fully sequenced. No alterations from the wild-type sequence, other than the intended single substitution cytosine → thymine at codon 173, were found (data not shown). The human apo A-II construct used in these studies has been previously described and uses the human apo A-I promoter in order to maximize apolipoprotein expression [11]. Both DNA constructs show a liver-specific expression [10,11]. The apo A-I_M and apo A-II constructs were co-injected into fertilized eggs from the FVB strain of mice. Founder animals carrying the two transgenes were crossed for several generations with C57Bl/6 mice which lacked murine apo A-I (A-IKO), due to gene targeting (provided by Dr N. Maeda) [9], to establish A-I_M/A-II/A-IKO transgenic lines. Control A-I/A-II/A-IKO mice were produced by multiple crosses between A-IKO mice and previously generated A-I/A-II mice [9,12]. The A-IKO littermates of the two transgenic lines were compared for the lipid/lipoprotein phenotype and no significant differences were observed. All the lipid and apolipoprotein analysis were performed on 12–16 week old mice of both sexes.

2.2. Plasma lipid, lipoprotein and apolipoprotein levels

Blood was collected after an overnight fast from the retroorbital plexus into tubes containing 0.1% (w/v) EDTA and centrifuged in a microcentrifuge for 10 min at 8000 rpm at 4°C. Plasma aliquots were immediately frozen at −20°C for apolipoprotein determinations by immunoturbidimetric assays, using a sheep antiserum against human apo A-I (Hoffmann La Roche, Basel, Switzerland) which also recognizes apo A-I_M [13], and an antiserum against human apo A-II (Boehringer, Mannheim, Germany). Lipid analyses were performed on fresh plasma stored at 4°C for less than 1 week. Cholesterol (total and unesterified) and triglycerides (TG) were measured by enzymatic methods (Roche) [14,15], HDL-cholesterol levels were measured after precipitation of apo B containing lipoproteins, as described [11].

2.3. Apolipoprotein distribution in transgenic mouse plasma

Aliquots of pooled plasma (500 μl) from five fasting A-I_M/A-II/A-IKO, A-I/A-II/A-IKO and A-IKO mice were adjusted to a density of 1.215 g/ml with solid KBr and centrifuged for 6 h at 4°C at 100 000 rpm in a Beckman TL 100 ultracentrifuge equipped with a Beckman TL 100.3 rotor. One microliter of the total lipoprotein fraction ($d < 1.215$ g/ml) was loaded onto a 7.5–17.5% SDS-polyacrylamide gradient gel and electrophoresed for 90 min at 100 V (Miniprotean apparatus, BioRad). When specified, disulfide bonds were cleaved by the addition of β-mercaptoethanol (β ME, 5% final concentration in sample buffer). After electrophoresis, proteins were transferred to nitrocellulose (Hybond C-EXTRA, Amersham, Buckinghamshire, UK) and immunodetected as described [16], using a monoclonal anti-human apo A-I antibody which also recognizes apo A-I_M (4HI, provided by Dr Y. Marcel) and a sheep anti-human apo A-II antiserum (Boehringer, Mannheim, Germany), both at 1:2000 dilution. Membranes were then incubated with rabbit anti-mouse and rabbit anti-sheep IgG conjugated with horse radish peroxidase (DAKO) at 1:2000 dilution, developed using the protocol and reagents of the ECL Western blotting detection system kit (Amersham, Buckinghamshire, UK) and exposed to Hyper-film ECL (Amersham film).

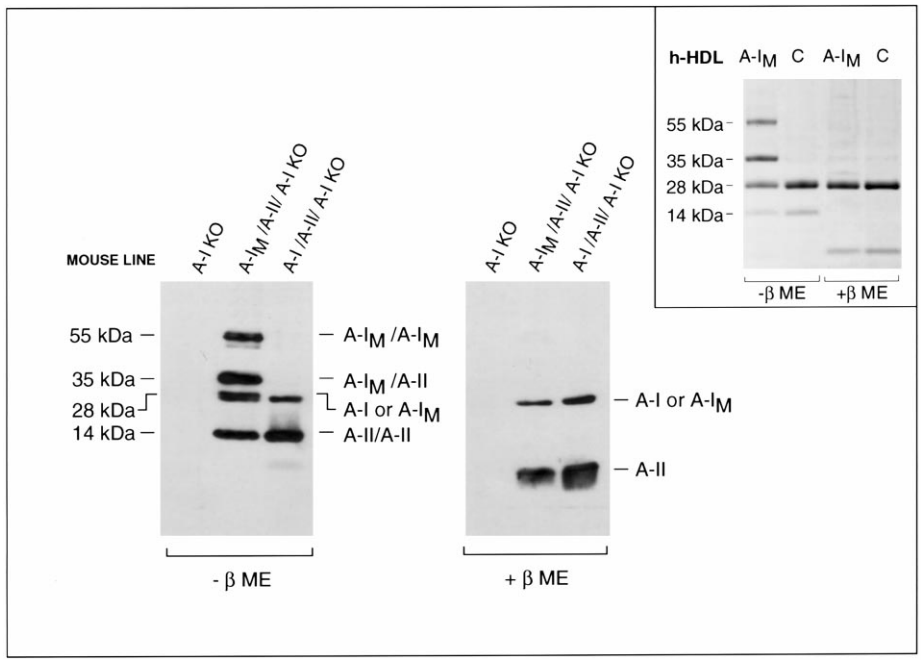


Fig. 1. Immunoblot analysis of human apolipoproteins in mouse plasma. Total lipoproteins from 500 μl of plasma, pooled from five fasted A-I_M/A-II/A-IKO, A-I/A-II/A-IKO and A-IKO mice, were separated by ultracentrifugation and loaded onto a 7.5–17.5% SDS-polyacrylamide gel under reducing and non-reducing conditions ($\pm\beta$ ME). Proteins were then transferred to nitrocellulose and immunodetected using specific anti-human apolipoproteins antibodies, as described in Section 2. The top right panel shows a Coomassie stained SDS-polyacrylamide gel of apo HDL from a normal subject (C) and an apo A-I_M carrier (A-I_M).

2.4. FPLC (fast protein liquid chromatography) analysis of plasma lipoproteins

Aliquots of plasma (500 μl) or total lipoproteins from three fasting A-I_M/A-II/A-IKO, A-I/A-II/A-IKO and A-IKO mice were applied to a Superose 6 10/30 column (Pharmacia) equilibrated with phosphate buffered saline, and eluted at a flow rate of 0.5 ml/min [17]. Twenty fractions (0.8 ml each) were collected. Cholesterol and TG contents were measured by enzymatic methods and human apolipoprotein distribution determined by immunoblotting as described above.

2.5. HDL subfraction distribution

Plasma samples (400 μl) were collected from three fasting A-I_M/A-II/A-IKO, A-I/A-II/A-IKO and A-IKO mice. HDL subfractions were separated by rate zonal ultracentrifugation [18] at 40 000 rpm for 23 h at 15°C in a Beckman L5–50 B ultracentrifuge equipped with a Beckman SW 41-Ti rotor; the tube effluent was monitored continuously at 280 nm. Fractions from major peaks were pooled and apolipoprotein distribution analyzed by immunoblotting, as described above.

To analyze the size distribution of mouse HDL, total lipoproteins ($d < 1.215$ g/ml), were loaded onto a non-denaturing 4–20% polyacrylamide gradient gel and electrophoresed for 18 h at 125 V at 4°C. Proteins were stained with Coomassie G-250 and HDL particle size was determined by densitometry (Ultroscan, LKB, Bromma, Sweden), using thyroglobulin (17.1 nm), ferritin (12.2 nm), catalase (10.4 nm), lactate dehydrogenase (8.15 nm) and albumin (7.1 nm) as protein standards [19].

2.6. Statistical analysis

Results are expressed as means $\pm$ SD. Group differences were tested by ANOVA for multiple comparisons, followed by Bonferroni's post-hoc test and considered to be statistically significant when $P < 0.05$.

3. Results

Mice expressing the human apo A-I, apo A-I_M and apo A-II transgenes were identified by immunoblot analysis of plasma apolipoproteins (Fig. 1). Under non-reducing conditions two bands were detectable in A-I/A-II/A-IKO transgenic mice, corresponding to apo A-I and to the dimeric form of human apo A-II (Fig. 1). Two additional bands, with apparent molecular weights of 35 and 55 kDa, and corresponding to the apo A-I_M/apo A-II heterodimer and apo A-I_M/apo A-I_M homodimer [13], were visible in A-I_M/A-II/A-IKO mice (Fig. 1). This pattern of bands resembles that observed in human plasma of apo A-I_M heterozygotes (Fig. 1, inset). However, in the transgenic mice which express

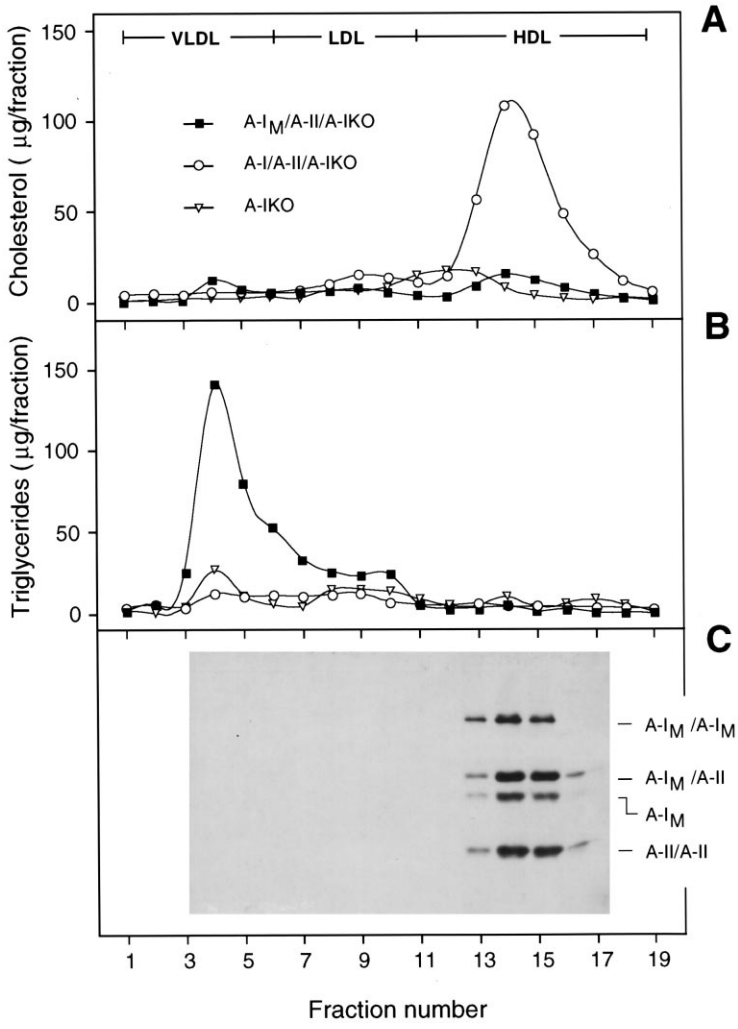


Fig. 2. FPLC analysis of mouse lipoproteins. Total lipoproteins from fasted A-I_M/A-II/A-IKO, A-I/A-II/A-IKO and A-IKO mice, were separated by ultracentrifugation and applied to a Superose 6B column. Fractions were collected and total cholesterol (A) and triglyceride (B) contents were measured. Aliquots of the A-I_M/A-II/A-IKO fractions were loaded onto a 7.5–17.5% SDS-polyacrylamide gel under non-reducing conditions and proteins were immunodetected (C) as described in Section 2.

apo A-I_M but not wild-type human apo A-I, the 28 kDa band represents exclusively apo A-I_M. Treatment of mouse and carriers' plasma with reducing agents cleaves the disulfide bonds in the dimeric proteins, which are no longer detectable (Fig. 1).

To determine the tissue distribution of human apolipoproteins expression in A-I_M/A-II/A-IKO mice, RNA blot analysis was performed using specific probes. The human apo A-I_M and apo A-II mRNA were detectable only in liver (data not shown).

Plasma apolipoprotein and lipid concentrations in the generated transgenic mice are shown in Table 1. The expression of human apo A-I and apo A-II genes in A-I/A-II/A-IKO mice resulted in elevated plasma apolipoprotein levels, similar to those previously reported for transgenic mice generated into a no A-IKO background [12]. In contrast, low plasma concentrations of apo A-I_M and apo A-II were found in A-I_M/A-II/A-IKO mice. Plasma total cholesterol concentrations were more than 3-fold greater in A-I/A-II/A-IKO than in A-IKO mice, the increase being mostly due to an increase in HDL-cholesterol levels (Table 1). In A-I_M/A-II/A-IKO mice, plasma total and HDL cholesterol levels were, instead, almost identical to those of A-IKO mice (Table 1). Mice expressing apo A-I_M were also characterized by moderately increased triglyceride levels, compared to both A-I/A-II/A-IKO and A-IKO mice. Moreover, an impaired cholesterol esterification was observed in A-I_M/A-II/A-IKO mice, as indicated by the significantly increased plasma unesterified to esterified cholesterol ratio (Table 1).

In order to analyze the lipid distribution among the different lipoprotein classes, mouse lipoproteins were separated by FPLC and the lipid content was measured on the collected fractions. The cholesterol profile of the three mouse lines confirmed the increase of the HDL

Table 1
Plasma lipid and apolipoprotein levels in transgenic mice

Mice	*n*	Total cholesterol (mg/dl)	Unesterified to esterified cholesterol	HDL cholesterol (mg/dl)	Triglycerides (mg/dl)	Human apo A-I (mg/dl)	Human apo A-I_M (mg/dl)	Human apo A-II (mg/dl)
A-I_M/A-II/A-IKO	12	33.4 ± 11.3^{b}	$0.57 \pm 0.09^{b, c}$	25.8 ± 13.0^{b}	$114.1 \pm 22.2^{b, c}$	—	68.3 ± 19.8	21.8 ± 5.3
A-I/A-II/A-IKO	13	99.7 ± 15.6	0.24 ± 0.04	85.4 ± 15.0	82.6 ± 8.9	206.8 ± 63.6	—	109.5 ± 10.5
A-IKO	25	31.4 ± 10.5	0.38 ± 0.11	23.1 ± 8.9	69.3 ± 12.6	—	—	—
Controls[a]	8	73.3 ± 17.4	0.42 ± 0.05	60.8 ± 16.9	70.7 ± 4.6	—	—	—

Results are expressed as means $\pm$ S.D.
[a] Data from age-sex matched C57Bl/6 mice.
[b] $P < 0.001$ vs. A-I/A-II/A-IKO mice.
[c] $P < 0.001$ vs. A-IKO mice.

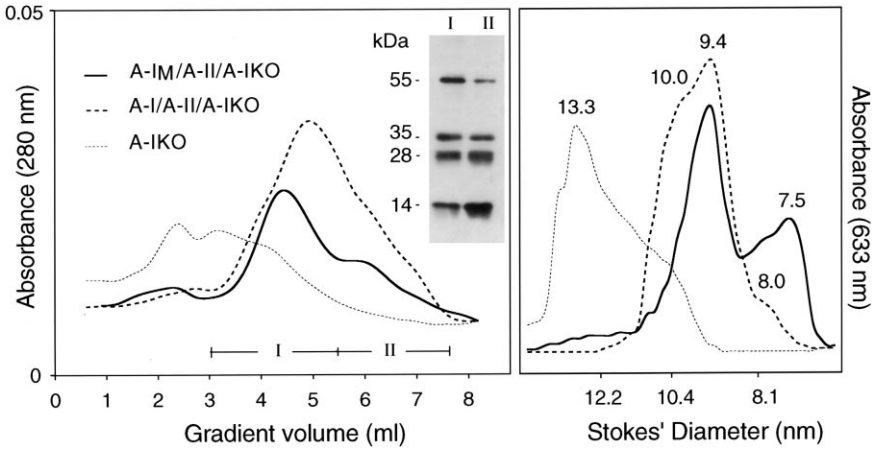


Fig. 3. Rate zonal ultracentrifugation (left) and nondenaturing GGE (right) profiles of HDL from fasted A-I_M/A-II/A-IKO, A-I/A-II/A-IKO and A-IKO mice. Left: fractions corresponding to fast floating (I) and slow floating (II). HDL from A-I_M/A-II/A-IKO mice were collected separately and aliquots were immunoblotted (inset), as described in Section 2. Right: lipoprotein fractions ($d < 1.215$ g/ml), separated by ultracentrifugation, were loaded onto a 4–30% nondenaturing polyacrylamide gel and proteins were stained with Coomassie G-250. HDL size was determined as described in Section 2.

fraction in A-I/A-II/A-IKO compared to A-IKO mice (Fig. 2A). Moreover, HDL from A-I/A-II/A-IKO mice eluted at a higher volume than A-IKO HDL, i.e. in a position similar to that of human HDL. In A-I_M/A-II/A-IKO mice, HDL particles eluted at the same position as A-I/A-II/A-IKO HDL, but were remarkably lower in concentration (Fig. 2A). A further difference in the lipoprotein cholesterol distribution among the three transgenic lines was observed in the VLDL fraction, which was significantly increased in A-I_M/A-II/A-IKO plasma. The analysis of the TG content in the FPLC fractions revealed that the increase of plasma TG observed in the A-I_M/A-II/A- IKO mice (Table 1), was essentially due to a marked increase in VLDL, without relevant differences in other lipoprotein fractions (Fig. 2B).

The distribution of the various apo A-I_M species in the lipoprotein fractions was investigated by SDS-polyacrylamide gel electrophoresis followed by immunoblot analysis. Both apo A-I_M/apo A-I_M and apo A-I_M/apo A-II homo- and heterodimers were exclusively present in the HDL size range (Fig. 2C).

The impact of apo A-I_M and apo A-II overexpression on HDL particle distribution was analyzed by rate zonal ultracentrifugation and nondenaturing gradient gel electrophoresis (GGE) (Fig. 3). The rate zonal HDL profile in A-I/A-II/A-IKO mice was characterized by a single heterogeneous peak of smaller and denser particles compared to A-IKO mice (Fig. 3, left panel). The HDL particles from A-I_M/A-II/A-IKO mice floated within the same density range of A-I/A-II/A-IKO HDL, but two distinct subfractions were detectable, i.e. a pattern closer to that of human HDL. These two subfractions were separately collected and analyzed for protein composition by immunoblot analysis (Fig. 3, left panel-inset). The apo A-I_M homodimer was present in higher amounts in the fast-floating, larger HDL (I) compared to denser particles (25 vs. 4%, respectively). In contrast, the apo A-II homodimer was more abundant in the denser, slow floating HDL (II) compared to the more buoyant, fast floating particles (40 vs. 22%, respectively).

The size characterization of HDL particles by nondenaturing GGE showed a predominance of very large particles (13.3 nm) in A-IKO mice, almost undetectable in mice expressing apo A-I or apo A-I_M (Fig. 3, right panel). HDL from A-I/A-II/A-IKO mice displayed a peak with a shoulder corresponding to particles of 9.4 and 10.0 nm diameter, respectively. A-I_M/A-II/A-IKO mice, on the other hand, showed two distinct peaks: one that corresponds to the size of the predominant A-I/A-II/A-IKO subpopulation (9.4 nm) and the other that is clearly of smaller size (7.5 nm).

4. Discussion

In this paper we have described the generation and the lipid/lipoprotein characterization of transgenic mice expressing human apo A-I_M and apo A-II, but no murine apo A-I. The expression of apo A-I_M and apo A-II resulted in the formation of homodimers apo A-I_M/apo A-I_M and heterodimers apo A-I_M/apo A-II in mouse plasma. A-I_M/A-II/A-IKO mice had low plasma cholesterol levels, with increased plasma unesterified to esterified cholesterol ratio, and increased TG, compared to mice expressing wild-type human apo A-I. Analysis of the lipoprotein profile of apo A-I_M mice revealed an increase in VLDL and a reduction in HDL levels. HDL particles of these mice were characterized by the pres-

ence of a small size subpopulation, not present in the other transgenic lines. All of those features reproduce the lipoprotein and apolipoprotein characteristics of humans with the apo A-I_M mutation.

The plasma HDL-cholesterol levels in A-I_M/A-II/A-IKO mice were much lower than those in A-I/A-II/A-IKO mice, and almost identical to the concentrations measured in the apo A-I deficient mice (Table 1). These low levels may be partially due to a low expression of the apo A-I_M transgene; indeed, the apo A-I_M mRNA levels in A-I_M/A-II/A-IKO mice were lower, with some animal to animal variability, compared to apo A-I mRNA levels in A-I/A-II/A-IKO mice (data not shown). In addition, the hypoalphalipoproteinemia in A-I_M/A-II/A-IKO mice could be consequent to an accelerated catabolism of the mutant protein, as observed in human carriers [20]. If occurring in transgenic mice, a rapid catabolism of apo A-I_M HDL may also contribute to the reduction of plasma apo A-II concentrations. However, the low HDL-cholesterol levels observed in apo A-I_M mice appear to be partly consequent to the biochemical features of the apo A-I variant, which allow the formation of cholesterol-poor HDL particles [7]. In fact, when A-I_M/A-II/A-IKO and A-I/A-II/A-IKO mice with similar plasma concentrations of apo A-I_M or apo A-I were compared (85 vs. 110 mg/dl), HDL-cholesterol levels in apo A-I_M mice were still more than 2-fold lower than those measured in apo A-I mice (data not shown).

The plasma unesterified to esterified cholesterol ratio (UC/EC) was significantly higher in A-I_M/A-II/A-IKO than in A-I/A-II/A-IKO and A-IKO mice (Table 1), i.e. similar to that observed in apo A-I_M carriers, who display an increased UC/EC (0.52) compared to non affected subjects (0.39). In the carriers, the increased UC/EC is mostly due to decreased (up to 50%) plasma LCAT concentrations [21]. Preliminary results in A-I_M/A-II/A-IKO mice, instead, showed only a 25% reduction of plasma LCAT activity, assessed by the exogenous substrate method [22], which reflects the plasma LCAT content. Thus, it is unlikely that this modest reduction in LCAT is the only mechanism responsible for the impaired cholesterol esterification in A-I_M/A-II/A-IKO mice. The elevated UC/EC may be due to a reduced ability of apo A-I_M to activate LCAT. Indeed, it has been recently shown that reconstituted particles containing apo A-I_M dimers are 30–70% less effective in activating LCAT, compared to particles containing wild-type human apo A-I [23]. Other potential mechanisms involve an accelerated 'selective uptake' of HDL cholesteryl esters by the liver, or an increased removal of UC from peripheral tissues. The availability of apo A-I_M transgenic mice will allow us to investigate these possibilities.

The elevation of plasma TG represents a consistent and reproducible finding in the A-I_M/A-II/A-IKO mice. The increased TG in transgenic mice is similar to that observed in apo A-I_M carriers, who also display moderately increased plasma TG levels compared to non affected subjects [24]. The underlying mechanism(s) by which apo A-I_M expression increases plasma TG is unknown. Studies in human carriers suggest that the low HDL levels and reduced EC formation could impair the cholesteryl ester transfer protein (CETP)-mediated exchange of EC for TG between HDL and VLDL, resulting in the accumulation of TG in VLDL [7]. However, mice lack CETP and therefore apo A-I_M must influence plasma TG metabolism by other mechanisms. The hypertriglyceridemia in the carriers does not appear to be caused by a reduced lipase activity [25], but the relative enrichment of VLDL with apo C-III may impair lipolysis and possibly delay the removal of these particles from plasma [26]. However, this awaits further investigation.

The HDL particle distribution, as analyzed by rate zonal ultracentrifugation, reveals the presence in A-I_M/A-II/A-IKO mice of two HDL subclasses differing by size, density and human apolipoprotein composition. The large HDL particles are enriched in apo A-I_M/apo A-I_M, as found in the human carriers [7]. Nondenaturing GGE of A-I/A-II/A-IKO and A-I_M/A-II/A-IKO HDL is consistent with the rate zonal profiles. A-I_M/A-II/A-IKO mouse HDL are characterized by the presence of a distinct subpopulation of small particles, not found in the other mouse lines. In human carriers, HDL are also characterized by a relative abundance of the smaller HDL_3 subpopulations [7]. Therefore, the expression of apo A-I_M in transgenic mice appears to mimic the human condition, in which dimerization of apo A-I_M restricts HDL particle size.

This study demonstrates that the expression of the human apo A-I_M mutation in a transgenic mouse also expressing human apo A-II generates a phenotype that is strikingly similar to that observed in human carriers, thus suggesting that the major lipid/lipoprotein abnormalities are directly consequent to the apo A-I_M mutation. Therefore, apo A-I_M transgenic mice appear to be a suitable model for the study of the impact of apo A-I_M in lipoprotein physiology. The question whether apo A-I_M can truly protect against the development of atherosclerosis [5] can now be addressed in an animal system which lacks the wild-type apo A-I.

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